

Application of bacteriophage therapy in the treatment of children with acute tonsillitis

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Abstract

Context: In pediatrics, there is an increasing interest in the therapeutic efficacy of inhaled bacteriophages for treating infectious diseases of the upper respiratory tract in children.

Aim: To analyze the effectiveness of inhaled bacteriophage therapy in children with acute tonsillitis (AT).

Settings and Design: Clinical observation of patients was based on a randomized, controlled research method.

Methods and Material: A total of 212 sick children aged 4–15 years who had AT were examined. Bacteriophage therapy was carried out by nebulizer inhalation using the liquid polyvalent piobacteriophage. Patients were divided into two groups based on the treatment method. One group of patients received a course of bacteriophage therapy against the background of standard treatment.

Results: When patients sought medical help, during the general clinical examination, the following were noted: cervical lymphadenopathy (98.6%), hyperemia of the tonsils and hyperplasia (98.1%), against the background of hyperthermia. Bacteriological analysis of the pharynx mucus showed that the main causative agents of AT in children were the bacteria *Streptococcus pyogenes*, *Staphylococcus aureus*, *Streptococcus pneumoniae*, and *Haemophilus influenzae*. Patients who underwent inhaled bacteriophage therapy in the early stages of treatment experienced a 1.4-fold greater disappearance of objective signs of pathological processes in the tonsils against the backdrop of rapid improvement in the well-being of sick children. Moreover, the use of a combination of antibiotic therapy and bacteriophage therapy led to a decrease in the detection of pathogenic bacteria, and the effectiveness of drug treatment increased four times.

Conclusion: The use of bacteriophage therapy in the complex treatment of AT in children contributes to a more rapid relief of the acute period against the background of an improvement in the subjective assessment of well-being by up to 25%, and a decrease in the detection of pathogenic bacteria.

Keywords: Acute tonsillitis, bacterium, bacteriophage, children, tonsils

KEY MESSAGES:

Based on bacteriological studies and assessment of the well-being of patients, the clinical effectiveness of bacteriophage therapy in the treatment of children with acute tonsillitis.

INTRODUCTION

In recent years, the incidence of respiratory diseases in children has increased, in particular, in the fall of 2023, in many countries worldwide, as well as in the countries of Central Asia. At the same time, the development of tactics for the treatment of acute respiratory diseases in children of different ages, against the background of the addition or complications of secondary bacterial infection, remains one of the questions widely studied in modern pediatrics, and provokes much discussion.^[1] One of these diseases is acute

tonsillitis (AT). Approximately 5%–15% of requests for medical consultations involve sick children suffering from AT.^[2] The main causative agent of AT is traditionally believed to be viruses.^[3] Then, pathogenic bacteria join them, or the impact of opportunistic bacteria, which are part of the oral microbiota, increases. Among bacteria, the most important etiological agent of AT is group A β -hemolytic *Streptococcus* [GABHS], and other bacteria play a decisive role in the clinical course of the disease.^[4,5] The

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development of new approaches for the treatment and improvement of these children is relevant. In pediatrics, scientific research is being conducted to study ways of non-invasive and highly effective methods of using medicines in children. One of these methods is the use of a nebulizer in the treatment of respiratory diseases. Administration of medications using a nebulizer is a generally accessible, easy-to-implement, and highly effective method. In addition, bacterial flora is one of the main premorbid backgrounds causing chronic childhood diseases.^[2] Over the past decade, pediatricians and infectious disease specialists have shown great interest in studying the therapeutic effect of bacteriophages in childhood infectious diseases. Moreover, the effect of bacteriophage therapy on the clinical course of AT in pediatric patients has not been fully studied. Studying the effectiveness of nebulized administration of bacteriophage to patients opens up broad opportunities in pediatric pulmonology. We analyzed the effectiveness of using inhaled bacteriophage therapy in children with AT together with traditional methods of treatment to study the effect of bacteriophage therapy on the effectiveness of treatment.

Ethical Review

To conduct clinical and laboratory studies on children in an outpatient setting, approval was received from the Ethics Committee of the Ministry of Health of the Republic of Uzbekistan (protocol no. 1/10-1484 dated February 27, 2021). The ethical standards are in accordance with the guidelines provided by the CPCSEA and World Medical Association Declaration of Helsinki on Ethical Principles for Medical Research involving humans for studies. Research register on the site ClinicalTrials.gov PRS <https://clinicaltrials.gov/ct2/show/NCT04682964> (IDENTIFIER: NCT04682964).

Patient Consent Form

Before conducting a clinical study, the patient's parents (or guardians) were informed about the examination being conducted, and the essence of the clinical study was explained. After the conversation, written permission was taken from parents (or guardians) to use the bacteriophage in the treatment of patients. Patient data and a list of clinical studies were indicated in the "Patient Consent Form." At the first request of the parents (or guardians), the use of bacteriophage during treatment was stopped. All examination results were simultaneously recorded in the patient's medical records.

SUBJECTS AND METHODS

Study Design

Clinical observation of patients was based on a randomized, controlled research method. The research design included several stages: initial, exploratory, and final. At the initial stage, patient selection criteria were developed, and groups of patients were created. At the research stage, a clinical and

laboratory study of children and adolescents was carried out on an outpatient basis in collaboration with specialized pediatricians (pediatric otolaryngologists, neurologists, pulmonologists, and infectious disease specialists). At the final stage, clinical data were analyzed, and the study results were statistically analyzed.

Eligibility Criteria

The inclusion criteria were as follows:

- children and adolescents aged 4–15 years with AT (study group) of both sexes.
The choice of this age group was due to the opportunity to objectively and subjectively assess the condition of sick children. In addition, at this age, children often develop AT as a separate type of pathology;
- patients who have the opportunity to visit the clinic;
- patients who do not have complications of AT (abscesses, rheumatism, scarlet fever, acute glomerulonephritis, and others).

The exclusion criteria were as follows:

- children under 4 years old and adolescents older than 15 years old;
- presence of chronic somatic diseases;
- the presence of psychosomatic and neurological disorders.

In total, during the scientific study, 212 children and adolescents (mean age 9.8 (2.37) years, $P \leq 0.01$) with AT were examined.

All the examined patients were divided into two identical groups according to age and sex. The first group included patients who were treated with AT during the disease course and who generally accepted standard treatment depending on the clinical variant of the pathology without bacteriophage therapy— $n=107$ (50.5% of 212) (for standard treatment, see below). The second group included patients who received liquid polyvalent pyobacteriophage (LPPB) in combination with standard traditional treatment— $n=105$ (49.5% of 212).

Research Methods

The patients received outpatient treatment at the clinic of the Tashkent Pediatric Medical Institute. All examined children underwent a general clinical examination (general blood and urine tests) and pharyngoscopy. Also, all patients underwent a bacteriological examination of a smear from the tonsils (culture of bacteria) according to standard methods.

Treatment Methods

Comprehensive outpatient treatment was based on standard clinical recommendations. Treatment included eliminating acute inflammatory processes in the tonsils and preventing general and local complications. Drug treatment included local treatment (gargling with antiseptic solutions: Furacilin, Betadine), antipyretic therapy, antihistamines,

antifungal agents (if there was a high risk of developing fungal pathology, then Fucin and Nystatin in age-related doses were prescribed), and vitamin therapy. Patients were prescribed complex antibacterial treatments, taking into account possible pathogens. The antibiotic of the first choice was amoxicillin at a dosage of 50 mg/kg/day twice a day, for 5 days.

The need for antibacterial treatment for AT is also indicated in the “Clinical practical guideline: tonsillitis I. Diagnosis And nonsurgical management,” noting a number of the benefits of antibacterial therapy.^[3]

Bacteriophage therapy was performed using the LPPB Sextaphage® (produced by Scientific Production Association Microgen HM RF Perm SPA Biomed, RU). LPPB was prescribed to be inhaled via a nebulizer, which irrigates the mucous membranes of the palatine tonsils. Inhalation was to be carried out once a day for a maximum of 10 minutes, starting on the first day of the disease, and continuing for 5 days. The total amount of LPPB to be used was 5 mL using a nebulizer.

The LPPB drug selection was based on the wide range of effects of the bacteriophage. In particular, the drug is capable of causing specific lysis of *Staphylococcus*, *Enterococcus*, *Streptococcus*, *Escherichia coli*, *Protea vulgaris*, *Pseudomonas aeruginosa*, *Klebsiella pneumoniae*, etc. The drug is recommended for use in treating inflammatory diseases of the otolaryngological organs, upper and lower respiratory tract, pleurisy, and other inflammatory diseases. A special feature of pyobacteriophages is that they do not affect other types of bacteria or disrupt the natural microflora. The given bacteriophage (Sextaphage®) we selected covered all isolated organisms.

The Effectiveness of the Therapy Was Evaluated

The effectiveness of the therapy was evaluated based on subjective and objective criteria. The subjective criterion included the results of a patient survey using a 5-point well-being scale developed by us (children’s well-being with acute diseases of the nasopharynx [WADN]). We developed this scale in collaboration with pediatric otolaryngologists during a 5-year study. The validity and reliability have been confirmed, and approval for the use of the scale has been received from the Academic Council of the Tashkent Pediatric Medical Institute. We developed this scale during clinical observation and conversations with sick children. The evaluation was carried out according to the following criteria: general (sleep quality, appetite, mood, and headache) or pathological (swallowing, sore throat, and dryness). The patients’ conditions were assessed daily at a certain time of day by the patients themselves or by their parents (guardians). The objective criteria for assessing the effectiveness of bacteriophage therapy were based on clinical examination (on the first day, the 3rd, and 6th days of treatment), laboratory (general blood test and bacterial culture of a tonsil smear) examination. A bacteriological

examination was carried out on the first day and the 6th day of treatment. No other bacterioscopic studies were performed.

Statistical Analysis

Based on the results obtained, a statistical analysis was carried out. Statistical processing of the results of the clinical laboratory study was carried out using a standard medical statistical program based on Excel (Microsoft Office 2013). This program was flexible and suitable for all types of statistical calculations used in biomedicine. Simultaneously, the sample size was calculated using this program. We calculated the arithmetic mean (M), the mean value, as a measure of central tendency in descriptive statistics of quantitative data, “ $M \pm S$,” where M was the mean, and S was the standard deviation (standard deviation, equal to the square root of the variance). Standard deviation was intended to describe normally distributed samples. The confidence interval was 95%. Comparisons between groups were made using the average population size for the study period. The results obtained are statistically significant at $P \leq 0.05$.

RESULTS

Clinical Manifestations of the Disease Before Treatment

The following are the results of a study on the clinical manifestations of a disease before treatment. During collecting an anamnesis of the onset of the disease (morbidity history), it was discovered that the occurrence of AT had a strong seasonality, mainly during the winter–spring period of the year. In particular, the occurrence of AT in winter was observed in 42% ($n = 89$) of patients; in spring 34% ($n = 72$) of patients; and in autumn 20.2% ($n = 43$) of patients.

During the study conducted in collaboration with a pediatric otolaryngologist, the following forms of AT were identified depending on the stage of pathology and the appearance of inflamed tonsils: catarrhal, in 65.57% ($n = 139$) of the children; follicular, 22.64% ($n = 48$); and lacunar, in 11.79% ($n = 25$) of the children. With an ulcerative-membranous form of the pathology, children were sent for inpatient treatment to specialized departments of children’s medical institutions.

During an objective clinical analysis, on the day of the outpatient consultation, patients noted weakness ($n = 209$; 96.8%), loss of appetite ($n = 201$; 94.8%), and headache ($n = 196$; 92.5%). Moreover, all patients complained of periodic hyperthermia ($n = 212$; 100%). During examination the following were noted: cervical lymphadenopathy ($n = 208$; 98.6%), hyperemia of the tonsils ($n = 212$; 100%), hyperplasia ($n = 208$; 98.1%), and the occurrence of exudates ($n = 102$; 48.1%). During clinical and laboratory examinations, pronounced leukocytosis with neutrophilia was noted against the background of accelerated erythrocyte sedimentation rate (ESR).

Treatment

Treatment of children with AT was based on generally accepted clinical recommendations and was of a pronounced symptomatic nature with the inclusion of antibacterial therapy based on the results of a bacteriological examination. Based on clinical recommendations, the drug of choice for antibacterial treatment was amoxicillin at age-specific doses.

These patients were consulted simultaneously by a pediatric otolaryngologist. Considering the clinical course of the disease, the age of the patients, the time required to obtain the results of a bacteriological study (bacterial culture), and the high risk of spread of the inflammatory process to the lower respiratory tract, to increase the effectiveness of treatment, patients were prescribed bacteriophage therapy against the background of antibacterial therapy.

In accordance with the objectives of our study, patients in the second group ($n = 105$; 49.5% out of 212), as part of standard treatment, were included in the treatment of LPPB from the first days of the disease. The standard treatment involved inhaling 5 mL of the medication.

As studies had shown that during the treatment, a decrease in swelling of the tonsils after 2–3 days of treatment, as well as a decrease in general signs of intoxication were noted.

Moreover, on the 6th day of conservative treatment, in children in group I, astheno-vegetative disorders persisted in 38.3%, while in children in group II, this figure was 17.1%. These were mainly general weakness (34.6% and 15.2%), loss of appetite (26.2% and 14.3%), and persistence of periodic headaches (29% and 11.4%) ($P \leq 0.05$) [Table 1].

The data indicate that despite complex conservative outpatient treatment, general signs of intoxication

(asthenic-vegetation disorders) persist on the 6th day of treatment. Moreover, considering the determination of the pattern of anemia in peripheral blood analysis as a consequence of intoxication and malnutrition during AT, anemia itself exacerbates the course of asthenic-vegetation disorders in patients. Moreover, in patients receiving inhaled bacteriophage therapy, there were 2.2 times fewer patients with asthenovegetative disorders than in patients in group I. These conditions can be characterized by the rapid elimination of toxins from the body due to the local immunostimulating and antibacterial effects of bacteriophages.

While analyzing local changes, the persistence of cervical lymphadenopathy was noted in 41.1% and 36.2% of patients on day 6, respectively, indicating a prolonged period of recovery of lymphoid tissue; moreover, changes in local pathological conditions in group II patients recovered on average 1.4 times faster than those in group I patients.

Moreover, in 77.1% ($n = 81$) of patients in group II, the level of leukocytes in the blood returned to normal on the 6th day of disease, against the background of conservative treatment. Moreover, 63.6% ($n = 68$) of the patients in group I were positive for this indicator ($P \leq 0.05$).

After treatment, on the 6th day, a new bacteriological examination of a smear from the mucous membranes of the tonsils was carried out. As the results of the present study showed, contamination depended on the therapy administered, especially in patients in group I. The frequency of bacterial contamination by pathogenic flora of the mucous membranes of the tonsils was 80.4% ($n = 86$ of 107), which indicated the need for continued treatment with local bacteriostatic drugs because long-term administration of antibiotics was not recommended for

Table 1: Clinical Manifestations of Acute Tonsillitis in Children Under Conservative Treatment ($P < 0.05$).

Clinical manifestation	Before treatment		On the third day of treatment				On the sixth day of treatment			
			Group I ($n = 107$)		Group II ($n = 105$)		Group I ($n = 107$)		Group II ($n = 105$)	
	<i>n</i>	%	<i>n</i>	%	<i>n</i>	%	<i>n</i>	%	<i>n</i>	%
General clinical signs of AT										
Astheno-vegetative disorders: including	212	100.0	86	80.37	61	58.10	41	38.32	18	17.14
General weakness	209	98.6	81	75.70	69	65.71	37	34.58	16	15.24
Headache	196	92.5	77	71.96	53	50.48	31	28.97	12	11.43
Loss of appetite	201	94.8	79	73.83	61	58.10	28	26.17	15	14.29
Intermittent hyperthermia	212	100.0	78	72.90	58	55.24	19	17.76	8	7.62
Local demonstration of the AT										
Odynophagy	209	98.58	82	76.64	72	68.57	38	35.51	26	24.76
Swelling of the tonsils	189	89.15	69	64.49	59	56.19	31	28.97	24	22.86
Hyperemia of the tonsils	212	100.00	80	74.77	71	67.62	37	34.58	27	25.71
The appearance of exudates	102	48.11	31	28.97	19	18.10	11	10.28	7	6.67
Cervical lymphadenopathy	208	98.1	76	71.03	67	63.81	44	41.12	38	36.19

The level of statistical significance of treatment outcome indicators relative to pre-treatment indicators is below 0.05 (or probability below 5%. This is the indicator for both study groups.

pediatric patients. Moreover, 18.1% ($n = 19$ out of 105) of the patients in group II were positive for this indicator, which indicated the high effectiveness of inhaled bacteriophage therapy.

To subjectively assess the effectiveness of the therapy, the Children's Well-Being Assessment Scale (WADN) was used in patients with acute diseases of the nasopharynx. According to the results of the study, at admission, the WADN was 1.88 (0.375) points; on the 3rd day of treatment, this indicator was 2.54 (0.453) for group I and 3.53 (0.394) points for group II; and on the sixth day, these indicators were 3.93 (0.106) and 4.44 (0.172) points, respectively ($P \leq 0.05$). As studies based on a personal assessment of the well-being of patients had shown, on the 3rd day of treatment, the difference in the assessment of well-being was 25.1% greater than that in group I. This difference is of clinical importance for the treating physician, because the subjective state of sick children reflects their general well-being, as an indicator of the effectiveness of the therapy.

The data obtained showed that patients in group II experienced a significant improvement on the beginning of the third day of treatment due to a reduction in intoxication phenomena, and on the 6th day, they were gradually "caught up" by patients in group I (the difference on the 6th day was 14.9%), even if they were still "lagging behind." Therefore, the use of inhaled bacteriophage therapy for the treatment of children and adolescents with AT on an outpatient basis beginning in the first days of medical consultation contributes to the rapid achievement of therapeutic efficacy while improving the medical quality of life of children.

The Results of the Bacteriological Examination Before and After Treatment

In accordance with the objectives of the study, all patients underwent bacteriological examination of a smear from the mucous layer of the tonsils. The study identified the following groups of main opportunistic and pathogenic bacteria: *Streptococcus pyogenes* (56.1% of 212 analyses), *Staphylococcus aureus* (33.5%), *Streptococcus pneumoniae* (25.5%), *Haemophilus influenzae* (18.8%), *Haemophilus parainfluenzae* (10.9%), *Streptococcus anginosus* (9.4%), and *Streptococcus dysgalactiae ssp. equisimilis* (8.6%), *Streptococcus agalactiae* (3.1%), and *Moraxella catarrhalis* (3.1%). Moreover, when analyzing the results obtained, it was noted that these pathogens most often appeared in the form of joint colonization (in 87% of the patients). Moreover, *S. pyogenes*, together with *Staph. aureus*, and *S. pneumoniae* were the predominant species cultured from AT patients, and the streptococcal bacterial flora was the dominant species (81.7%). Considering the high pathogenicity of *S. pyogenes*, *S. aureus*, and *S. pneumoniae* were identified as indicators of the effectiveness of bacteriophage therapy.

As our studies have shown, after a course of antibiotic therapy, on the 6th day of conservative treatment, the

detection of *S. pyogenes* decreases from 53.3% to 4.7%, that of *S. aureus* decreases from 35.5% to 4.7%, and that of *S. pneumoniae* decreases from 26.2% to 3.7%, representing an average 8.7-fold decrease in the frequency of detection of these bacteria. Moreover, the use of a combination of antibiotic therapy and bacteriophage therapy on the 6th day of treatment leads to an average reduction of more than 30 times in the detectability of these bacteria. Therefore, after the first days of combined antibiotic and bacteriophage therapy, the effectiveness of drug treatment can increase 4-fold in patients with AT.

DISCUSSION

Clinical Manifestations of AT in Children

AT is one of the most common pathological otolaryngological conditions in children and adolescents.^[6] Due to the many variables involved in classifying the levels of evidence for different approaches to the treatment of AT, this disease required varied evaluation.^[3] If this need is met, therapeutic approaches should have a major impact in the years to come.^[7] The tonsils, which are important parts of the immune system, are involved in the acquisition of the immune system and contain T lymphocytes, macrophages, and B lymphocyte germinal centers.^[8] Moreover, the tonsils represent the first step in the lymphoid tissue system—the mucosa-associated lymphoid tissue system.^[4,7] Moreover, according to some authors, the main phase of acquired immunity lasts up to 6 years, and during this period, the palatine tonsils are physiologically hyperplastic; subsequently, at up to 12 years of age, involution occurs, manifested by tissue regression.^[5,9] Therefore, at this age, catarrhal manifestations of AT are characteristic. As our studies showed, in outpatient settings, the catarrhal form of AT was diagnosed in 65.63% ($n = 84$) of children; follicular, 22.66% ($n = 29$); and lacunar, 11.72% ($n = 15$) of children. These findings indicate the predominance of the catarrhal form of AT in outpatients.

During a clinical study, 98.6% ($n = 209$) of patients were diagnosed with odynophagia. Some authors believe that odynophagia for 24–48 hours is a manifestation of the prodromal period in viral upper respiratory tract infections.^[5] During clinical examination, hypertrophy of the pharyngeal tonsils (adenoids) was simultaneously noted. Hyperplasia of the pharyngeal tonsil developed from hypertrophy of the reaction centers of the lymphatic tissue.

As our studies showed, the main causative agent of AT in children was the bacterium *S. pyogenes*. (GABHS), *S. aureus*, and *H. influenzae* (from 24.2% to 42.2%). Its co-colonization with *M. catarrhalis* appears to increase the risk of developing AT. It should be noted that the invasion of *S. aureus* is influenced by various virulence factors, particularly adhesins or microbial "surface components," that recognize adhesive matrix molecules. Staphylococcal adhesion to host cells is often mediated by binding to bridging matrix molecules that

also bind to host cells through specific receptors, such as $\beta 1$ integrins. Along with the bacterial group, the presence of *Candida sp.* was determined—16.5% ($n=35$). The main causative agent of AT in children is GABHS, and this should be considered during treatment.^[10]

Additionally, Yeoh *et al.* studying the microbiota of the oral cavity indicate that in AT, group A *Streptococci* (GABHS), *Staphylococcus* most often predominate *S. aureus*, *S. pneumoniae*, *H. influenzae*, and representatives of the group *Prevotella*, *Bacteroides*, *Fusobacterium*, family *Porphyromonas* and *Veillonella*, which simultaneously cause acute diseases of the upper respiratory tract in combination.^[11,12] These authors recommend that antibacterial treatment be directed specifically against these pathogens. Moreover, some countries have started to recommend the use of phage therapy as an alternative to antibiotic therapy.^[13]

At the same time, according to Büttner *et al.*,^[2] the use of antibacterial treatment for AT helps to reduce the duration of the patient's contagiousness (after approximately 24 hours, patients receiving antibiotic treatment are no longer contagious); relieve symptoms of tonsillitis, pharyngitis and fever; reduce purulent complications; and prevent secondary immunogenic diseases, such as rheumatic fever or acute poststreptococcal glomerulonephritis. However, these arguments remain debatable.^[2]

Use of Bacteriophage Therapy

In recent years, there has been growing interest in the use of bacteriophages in pediatrics. There are numerous documented data indicating the effectiveness of the use of bacteriophages in the treatment of respiratory tract diseases in children.^[12,14] Moreover, given the annual spread of COVID-19, these patients also began to use inhalation phage therapy.^[15]

According to our findings, multifactorial development occurs when secretion stagnates due to inflammatory processes and tissue hyperplasia, leading to further inflammation. Therefore, in these patients, viruses cause immune and inflammatory reactions in the upper respiratory tract, and bacterial superinfection leads to the development of AT.

In addition, in 2019, vaccines were developed against GABHS, which could subsequently prevent the development of acute infectious diseases of the upper respiratory tract caused by GABHS pathogens; however, these studies were still in the experimental stage.^[14]

During the treatment, we noted a rapid improvement in the condition of the sick children in the second group of patients. In particular, compared to that in the first group, clinical improvement in the condition of patients was observed 1.4 times faster in children receiving bacteriophage treatment.

The use of nebulization for inhaled bacteriophage therapy promotes deeper penetration into the tissues of not only the

upper but also the lower respiratory tract, in contrast to the oral use of bacteriophages.^[16,17] Moreover, the use of inhaled bacteriophage therapy is becoming a current trend in pediatrics due to the noninvasive nature and availability of this method for treating respiratory diseases in children.^[18,19]

Therefore, at the same time, the occurrence of acute inflammatory diseases caused by pathogenic bacterial microflora is prevented. This fact has been proven by our research.

Based on these results, we can conclude that the use of bacteriophages in the outpatient treatment of children with AT has good therapeutic efficacy.

CONCLUSION

The pathogenesis of AT in children involves *S. pyogenes*, *S. aureus*, *S. pneumoniae*, *H. influenza*, and *H. parainfluenzae*, which occur in variable relative proportions. The use of inhaled bacteriophage therapy in the conservative treatment of AT in children in the early stages of treatment contributes to 1.4 times faster relief of the acute period, thereby providing improvement in subjective assessment of well-being 25% faster than in the comparison group. Therefore, the use of inhaled bacteriophage therapy on an outpatient basis allows one to achieve the fastest therapeutic effect and prevent bacterial complications in the lower respiratory tract.

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Conflicts of interest

There are no conflicts of interest.

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